Dong Su Kim

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WORK EXPERIENCE

Conboy Lab **Sept. 2023 -- Oct. 2024**

Machine Learning Research Associate Berkeley, CA

- Designed a distributed scalable data pipeline for multi-terabyte biological datasets, improving preprocessing speed 30% through parallelization, optimized I/O, and data batching.
- Developed end-to-end ML workflows including feature extraction, model training, evaluation, and automated retraining hooks.
- Productionized ML pipelines using Docker, CI/CD, and versioned experiment tracking—enabling reproducible execution across cloud and HPC compute environments.
- Implemented model-serving infrastructure using FastAPI + AWS (S3, EC2), supporting async inference and multi-stage preprocessing.
- Built internal tools for data validation, metadata extraction, and model result aggregation, improving reliability across downstream teams.
- Collaborated cross-functionally with engineers and scientists to deploy ML models into automated pipelines and improve operational performance.

PROJECTS

Sequence Modeling & ML Inference Pipeline

Python, PyTorch, FastAPI, Postgres, Docker, AWS

End-to-end system for protein sequence inference.

- Built a **microservice pipeline** with preprocessing, embedding generation (BERT/ESM2), model inference, and results aggregation.
- Integrated **Postgres + S3** for efficient storage and retrieval of sequence data.
- Deployed pipeline via Dockerized services on **EC2**; implemented monitoring for throughput and error rates.
- Improved predictive performance **15%** through custom augmentation and hyperparameter tuning.

Hunger

Python, Java, CSS, HTML, FastAPI, PyTorch, Google Places API

Contextually Scalable Restaurant Rating Tracker - Beli Clone

- Built a **Full-stack** web application with a backend powered by **FastAPI** and a frontend in **Javascript** with no runtime external dependencies and complete with a powerful **geocoding** implementation.
- Designed a **machine learning** guided **recommendation system** with an interactive questionnaire that narrows hundreds of restaurants down to 1–3 matches via heuristics and a **XGBoost** model.
- Developed a synthetic tag and user persona generator with weighted cuisine-category distributions to simulate realistic user preference data for downstream ML evaluation.
- Engineered highly modular async-await style FastAPI endpoints with reusable components and client-side validation with an extensive design doc detailing technical implementations, justifications, and function descriptions.

EDUCATION

UC Berkeley

Computer Science, Bioengineering Berkeley, CA

- Relevant Coursework: Machine Learning, AI, Algorithms, Data Structures, Computer Architecture, Computer Vision

SKILLS

Programming: Python, SQL, Java, C, JavaScript/TypeScript
ML & DL: PyTorch, TensorFlow, Keras, Transformers, EfficientNet, Hugging Face
ML Engineering: Model deployment, experiment tracking, distributed data processing, evaluation pipelines, GPU optimization
Data & Infra: Postgres, FastAPI, Docker, AWS (EC2, S3), CI/CD, Linux, HPC